

* Rank 0 tensor. * Rank 1 tensor.

$$\begin{bmatrix} T_x & 0 & T_z \end{bmatrix}$$

$$T_x = \mathbb{T}_x^{(1)}$$

$$T_z = \mathbb{T}_z^{(1)}$$

* Rank 2 tensor (s).

$$\begin{bmatrix} 0 & T_{xy} & 0 \\ T_{xy} & 0 & T_{yz} \\ 0 & T_{yz} & 0 \end{bmatrix}$$

$$T_{xy} = \mathbb{T}_{xy}^{(1)}$$

$$T_{yz} = \mathbb{T}_{yz}^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & T_{xy} & 0 \\ -T_{xy} & 0 & T_{yz} \\ 0 & -T_{yz} & 0 \end{bmatrix}$$

$$T_{xy} = \mathbb{M}_z^{(1)}$$

$$T_{yz} = \mathbb{M}_x^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} T_{xxx} & 0 & T_{xxz} \\ T_{yyx} & 0 & T_{yyz} \\ T_{zzx} & 0 & T_{zzz} \\ 0 & T_{yyz} & 0 \\ T_{xxz} & 0 & T_{zzx} \\ 0 & T_{yyx} & 0 \end{bmatrix}$$

$$T_{xxx} = 3\mathbb{T}_x^{(1)} + 2\mathbb{T}_x^{\alpha(1)}$$

$$T_{xxz} = \mathbb{T}_z^{(1)} - \mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\beta(1)}$$

$$T_{yyx} = \mathbb{T}_x^{(1)} - \mathbb{T}_x^{\alpha(1)} + \mathbb{T}_x^{\beta(1)}$$

$$T_{yyz} = \mathbb{T}_z^{(1)} - \mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\beta(1)}$$

$$T_{zzx} = \mathbb{T}_x^{(1)} - \mathbb{T}_x^{\alpha(1)} - \mathbb{T}_x^{\beta(1)}$$

$$T_{zzz} = 3\mathbb{T}_z^{(1)} + 2\mathbb{T}_z^{\alpha(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & T_{xxz} \\ T_{yyx} & 0 & T_{yyz} \\ T_{zzx} & 0 & 0 \\ 0 & -\frac{T_{yyz}}{2} & 0 \\ -\frac{T_{xxz}}{2} & 0 & -\frac{T_{zzx}}{2} \\ 0 & -\frac{T_{yyx}}{2} & 0 \end{bmatrix}$$

$$T_{xxz} = 2\mathbb{M}_{xy}^{(1)} + 2\mathbb{T}_z^{(2)}$$

$$T_{yyx} = 2\mathbb{M}_{yz}^{(1)} + 2\mathbb{T}_x^{(2)}$$

$$T_{yyz} = -2\mathbb{M}_{xy}^{(1)} + 2\mathbb{T}_z^{(2)}$$

$$T_{zzx} = -2\mathbb{M}_{yz}^{(1)} + 2\mathbb{T}_x^{(2)}$$

$$\frac{T_{yyy}}{2} = -\mathbb{M}_{xy}^{(1)} + \mathbb{T}_z^{(2)}$$

$$\frac{T_{xxz}}{2} = \mathbb{M}_{xy}^{(1)} + \mathbb{T}_z^{(2)}$$

$$\frac{T_{zzx}}{2} = -\mathbb{M}_{yz}^{(1)} + \mathbb{T}_x^{(2)}$$

$$\frac{T_{yyx}}{2} = \mathbb{M}_{yz}^{(1)} + \mathbb{T}_x^{(2)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} 0 & T_{yzy} & 0 \\ T_{zxx} & 0 & T_{zzz} \\ 0 & T_{xyy} & 0 \end{bmatrix}$$

$$T_{yzy} = \mathbb{M}_{xy}^{(2)} + \mathbb{T}_z^{(3)}$$

$$T_{zxx} = \mathbb{M}_{xy}^{(2)} - \mathbb{T}_z^{(3)}$$

$$T_{zzz} = \mathbb{M}_{yz}^{(2)} + \mathbb{T}_x^{(3)}$$

$$T_{xyy} = \mathbb{M}_{yz}^{(2)} - \mathbb{T}_x^{(3)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & T_{xyz} & 0 & T_{xxx} \\ 0 & 0 & 0 & T_{yyz} & 0 & T_{yyx} \\ 0 & 0 & 0 & T_{zzz} & 0 & T_{zzx} \\ T_{xyz} & T_{yyz} & T_{zzz} & 0 & T_{yzz} & 0 \\ 0 & 0 & 0 & T_{yzz} & 0 & T_{zxx} \\ T_{xxx} & T_{yyx} & T_{zzx} & 0 & T_{zxy} & 0 \end{bmatrix}$$

$$T_{xxy} = 2\mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)}$$

$$T_{xxx} = \mathbb{T}_{4z}^{\alpha(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} - \mathbb{T}_z^{\beta(1)}$$

$$T_{yyy} = \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)}$$

$$T_{yyx} = -\mathbb{T}_{4z}^{\alpha(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} - \mathbb{T}_z^{\beta(1)}$$

$$T_{zzz} = -\mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)}$$

$$T_{zzx} = \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_z^{\beta(1)}$$

$$T_{yzz} = \mathbb{T}_{xy}^{(2)} + 2\mathbb{T}_z^{\beta(1)}$$

$$T_{zxy} = 2\mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(2)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & T_{xyz} & 0 & T_{xxy} \\ 0 & 0 & 0 & T_{yyz} & 0 & T_{yyx} \\ 0 & 0 & 0 & T_{zzz} & 0 & T_{zzx} \\ -T_{xyz} & -T_{yyz} & -T_{zzz} & 0 & T_{yzz} & 0 \\ 0 & 0 & 0 & -T_{yzz} & 0 & T_{zxx} \\ -T_{xxy} & -T_{yyx} & -T_{zzx} & 0 & -T_{zxy} & 0 \end{bmatrix}$$

$$\begin{aligned}
T_{xxyz} &= -4\mathbb{M}_x^{\beta(1)} + \mathbb{T}_{yz}^{(3)} \\
T_{xxxy} &= 2\mathbb{M}_z^{(1)} - 2\mathbb{M}_z^{\alpha(1)} + 2\mathbb{M}_z^{\beta(1)} + \mathbb{T}_{xy}^{(3)} \\
T_{yyyz} &= 2\mathbb{M}_x^{(1)} - 2\mathbb{M}_x^{\alpha(1)} + 2\mathbb{M}_x^{\beta(1)} + \mathbb{T}_{yz}^{(3)} \\
T_{yyxy} &= -2\mathbb{M}_z^{(1)} + 2\mathbb{M}_z^{\alpha(1)} + 2\mathbb{M}_z^{\beta(1)} + \mathbb{T}_{xy}^{(3)} \\
T_{zzyz} &= -2\mathbb{M}_x^{(1)} + 2\mathbb{M}_x^{\alpha(1)} + 2\mathbb{M}_x^{\beta(1)} + \mathbb{T}_{yz}^{(3)} \\
T_{zzxy} &= -4\mathbb{M}_z^{\beta(1)} + \mathbb{T}_{xy}^{(3)} \\
T_{yzzx} &= -\mathbb{M}_z^{(1)} - 4\mathbb{M}_z^{\alpha(1)} \\
T_{zxxxy} &= -\mathbb{M}_x^{(1)} - 4\mathbb{M}_x^{\alpha(1)}
\end{aligned}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} 0 & T_{yzzx} & 0 \\ T_{yzzx} & 0 & T_{zxxxy} \\ 0 & T_{zxxxy} & 0 \end{bmatrix}$$

$$\begin{aligned}
T_{yzzx} &= 2\mathbb{T}_{xy}^{(4)} \\
T_{zxxxy} &= 2\mathbb{T}_{yz}^{(4)}
\end{aligned}$$

* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & T_{yzzx} & 0 \\ -T_{yzzx} & 0 & T_{zxxxy} \\ 0 & -T_{zxxxy} & 0 \end{bmatrix}$$

$$\begin{aligned}
T_{yzzx} &= -\mathbb{M}_z^{(2)} \\
T_{zxxxy} &= -\mathbb{M}_x^{(2)}
\end{aligned}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} T_{xxyz} & 0 & T_{xxxy} \\ T_{yyyz} & 0 & T_{yyxy} \\ T_{zzyz} & 0 & T_{zzxy} \\ 0 & T_{yzzx} & 0 \\ T_{zxxy} & 0 & T_{zxxxy} \\ 0 & T_{xyzx} & 0 \end{bmatrix}$$

$$\begin{aligned}
T_{xxyz} &= \mathbb{M}_x^{(3)} + 2\mathbb{M}_x^{(4)} + 2\mathbb{M}_x^{\alpha(2)} \\
T_{xxxy} &= \mathbb{M}_z^{(3)} - \mathbb{M}_z^{\alpha(2)} + \mathbb{M}_z^{\beta(2)} - 2\mathbb{T}_{xy}^{(5)} \\
T_{yyyz} &= \mathbb{M}_x^{(3)} - \mathbb{M}_x^{\alpha(2)} + \mathbb{M}_x^{\beta(2)} - 2\mathbb{T}_{yz}^{(5)} \\
T_{yyxy} &= \mathbb{M}_z^{(3)} - \mathbb{M}_z^{\alpha(2)} - \mathbb{M}_z^{\beta(2)} + 2\mathbb{T}_{xy}^{(5)} \\
T_{zzyz} &= \mathbb{M}_x^{(3)} - \mathbb{M}_x^{\alpha(2)} - \mathbb{M}_x^{\beta(2)} + 2\mathbb{T}_{yz}^{(5)} \\
T_{zzxy} &= \mathbb{M}_z^{(3)} + 2\mathbb{M}_z^{(4)} + 2\mathbb{M}_z^{\alpha(2)} \\
T_{yzzx} &= \mathbb{M}_z^{(4)} - \mathbb{M}_z^{\alpha(2)} - \mathbb{M}_z^{\beta(2)} - \mathbb{T}_{xy}^{(5)} \\
T_{zxxy} &= \mathbb{M}_z^{(4)} - \mathbb{M}_z^{\alpha(2)} + \mathbb{M}_z^{\beta(2)} + \mathbb{T}_{xy}^{(5)} \\
T_{zxxxy} &= \mathbb{M}_x^{(4)} - \mathbb{M}_x^{\alpha(2)} - \mathbb{M}_x^{\beta(2)} - \mathbb{T}_{yz}^{(5)} \\
T_{xyzx} &= \mathbb{M}_x^{(4)} - \mathbb{M}_x^{\alpha(2)} + \mathbb{M}_x^{\beta(2)} + \mathbb{T}_{yz}^{(5)}
\end{aligned}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} T_{yzxx} & T_{zyyy} & T_{yzzz} & 0 & T_{yzzx} & 0 \\ 0 & 0 & 0 & T_{zxxy} & 0 & T_{zxxxy} \\ T_{xyxx} & T_{xyyy} & T_{xyzx} & 0 & T_{xyzx} & 0 \end{bmatrix}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} T_{yzxx} & T_{yzyy} & T_{yzzz} & 0 & T_{yzzx} & 0 & 0 & T_{yzxz} & 0 \\ 0 & 0 & 0 & T_{zxyz} & 0 & T_{zxxy} & T_{zxxz} & 0 & T_{zxyx} \\ T_{xyxx} & T_{xyyy} & T_{xyzx} & 0 & T_{xyzx} & 0 & 0 & T_{xyxz} & 0 \end{bmatrix}$$

$$\begin{aligned} T_{yzxx} &= \mathbb{M}_x^{(5)} + 2\mathbb{M}_x^{(6)} + 2\mathbb{M}_x^{\alpha(3)} \\ T_{yzyy} &= \mathbb{M}_x^{(5)} - \mathbb{M}_x^{\alpha(3)} + \mathbb{M}_x^{\beta(3)} - 2\mathbb{T}_{yz}^{(6)} \\ T_{yzzz} &= \mathbb{M}_x^{(5)} - \mathbb{M}_x^{\alpha(3)} - \mathbb{M}_x^{\beta(3)} + 2\mathbb{T}_{yz}^{(6)} \\ T_{yzzx} &= -\mathbb{M}_z^{(2)} + \mathbb{M}_z^{(6)} - \mathbb{M}_z^{\alpha(3)} + \mathbb{M}_z^{\beta(3)} + 2\mathbb{T}_{xy}^{(4)} + \mathbb{T}_{xy}^{(6)} \\ T_{yzxz} &= \mathbb{M}_z^{(2)} + \mathbb{M}_z^{(6)} - \mathbb{M}_z^{\alpha(3)} + \mathbb{M}_z^{\beta(3)} - 2\mathbb{T}_{xy}^{(4)} + \mathbb{T}_{xy}^{(6)} \\ T_{zxyz} &= \mathbb{M}_z^{(2)} + \mathbb{M}_z^{(6)} - \mathbb{M}_z^{\alpha(3)} - \mathbb{M}_z^{\beta(3)} + 2\mathbb{T}_{xy}^{(4)} - \mathbb{T}_{xy}^{(6)} \\ T_{zxxy} &= -\mathbb{M}_x^{(2)} + \mathbb{M}_x^{(6)} - \mathbb{M}_x^{\alpha(3)} + \mathbb{M}_x^{\beta(3)} + 2\mathbb{T}_{yz}^{(4)} + \mathbb{T}_{yz}^{(6)} \\ T_{zxxz} &= -\mathbb{M}_z^{(2)} + \mathbb{M}_z^{(6)} - \mathbb{M}_z^{\alpha(3)} - \mathbb{M}_z^{\beta(3)} - 2\mathbb{T}_{xy}^{(4)} - \mathbb{T}_{xy}^{(6)} \\ T_{zxyx} &= \mathbb{M}_x^{(2)} + \mathbb{M}_x^{(6)} - \mathbb{M}_x^{\alpha(3)} + \mathbb{M}_x^{\beta(3)} - 2\mathbb{T}_{yz}^{(4)} + \mathbb{T}_{yz}^{(6)} \\ T_{xyxx} &= \mathbb{M}_z^{(5)} - \mathbb{M}_z^{\alpha(3)} + \mathbb{M}_z^{\beta(3)} - 2\mathbb{T}_{xy}^{(6)} \\ T_{xyyy} &= \mathbb{M}_z^{(5)} - \mathbb{M}_z^{\alpha(3)} - \mathbb{M}_z^{\beta(3)} + 2\mathbb{T}_{xy}^{(6)} \\ T_{xyzx} &= \mathbb{M}_z^{(5)} + 2\mathbb{M}_z^{(6)} + 2\mathbb{M}_z^{\alpha(3)} \\ T_{xyzx} &= \mathbb{M}_x^{(2)} + \mathbb{M}_x^{(6)} - \mathbb{M}_x^{\alpha(3)} - \mathbb{M}_x^{\beta(3)} + 2\mathbb{T}_{yz}^{(4)} - \mathbb{T}_{yz}^{(6)} \\ T_{xyxz} &= -\mathbb{M}_x^{(2)} + \mathbb{M}_x^{(6)} - \mathbb{M}_x^{\alpha(3)} - \mathbb{M}_x^{\beta(3)} - 2\mathbb{T}_{yz}^{(4)} - \mathbb{T}_{yz}^{(6)} \end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & T_{xxxy} & 0 \\ T_{yyyyx} & 0 & T_{yyyyz} \\ 0 & T_{zzzy} & 0 \\ 0 & T_{yyzy} & 0 \\ 0 & T_{zzxy} & 0 \\ T_{xxyx} & 0 & T_{xxyz} \\ T_{yzzx} & 0 & T_{yzzz} \\ 0 & T_{zxxxy} & 0 \\ 0 & T_{xyyy} & 0 \\ -\frac{T_{yyyyz}}{2} + \frac{T_{yyzy}}{2} + T_{zxxxy} & 0 & \frac{T_{xxxy}}{2} - \frac{T_{xxyz}}{2} + T_{xyyy} - T_{yyyyx} + T_{zzxy} \end{bmatrix}$$

$$\begin{aligned} T_{xxyx} &= \mathbb{T}_{4z}^{\alpha(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} - 2\mathbb{T}_{xy}^{(5)} - \mathbb{T}_z^{\beta(1)} \\ T_{yyyyx} &= -\mathbb{T}_{4z}^{\alpha(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} - 2\mathbb{T}_{xy}^{(5)} - \mathbb{T}_z^{\beta(1)} \\ T_{yyyyz} &= \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} - 2\mathbb{T}_{yz}^{(5)} \\ T_{zzzy} &= -\mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} - 2\mathbb{T}_{yz}^{(5)} \\ T_{yyzy} &= \mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} + \mathbb{T}_{yz}^{(3)} + 2\mathbb{T}_{yz}^{(5)} \\ T_{zxxxy} &= \mathbb{T}_{xy}^{(1)} + \mathbb{T}_{xy}^{(3)} + 2\mathbb{T}_z^{\beta(1)} \\ T_{xxyx} &= \mathbb{T}_{4z}^{\alpha(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} + 2\mathbb{T}_{xy}^{(5)} - \mathbb{T}_z^{\beta(1)} \\ T_{xxyz} &= 2\mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + \mathbb{T}_{yz}^{(3)} \\ T_{yzzx} &= \mathbb{T}_{xy}^{(2)} - \mathbb{T}_{xy}^{(5)} + 2\mathbb{T}_z^{\beta(1)} \\ T_{yzzz} &= -\mathbb{T}_{4x}^{\alpha(1)} - \mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(1)} + 2\mathbb{T}_{yz}^{(2)} - \mathbb{T}_{yz}^{(3)} \\ T_{zxxxy} &= 2\mathbb{T}_x^{\beta(1)} + \mathbb{T}_{yz}^{(2)} - \mathbb{T}_{yz}^{(5)} \\ T_{xyyy} &= -\mathbb{T}_{4z}^{\alpha(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} - \mathbb{T}_{xy}^{(3)} - \mathbb{T}_z^{\beta(1)} \end{aligned}$$